

Cassandra Experiments and Repetition Counts

One completed experiment run: 311 trials in total — 300 main trials, 10 read-repair trials, and one timestamp-control trial.

Measured run: 20260912T063741Z. Each number in the table below is the number of repetitions performed. Configuration names mean write consistency / read consistency.

Scenario	Write / read	RYW	MR	MW	WFR	Total
Normal operation	ONE/ONE	5	5	5	5	20
Normal operation	QUORUM/ONE	5	5	5	5	20
Normal operation	ONE/QUORUM	5	5	5	5	20
Normal operation	QUORUM/QUORUM	5	5	5	5	20
Normal operation	ALL/ALL	5	5	5	5	20
One node killed	ONE/ONE	5	5	5	5	20
One node killed	QUORUM/ONE	5	5	5	5	20
One node killed	ONE/QUORUM	5	5	5	5	20
One node killed	QUORUM/QUORUM	5	5	5	5	20
One node killed	ALL/ALL	5	5	5	5	20
Network partition	ONE/ONE	5	5	5	5	20
Network partition	QUORUM/ONE	5	5	5	5	20
Network partition	ONE/QUORUM	5	5	5	5	20
Network partition	QUORUM/QUORUM	5	5	5	5	20
Network partition	ALL/ALL	5	5	5	5	20
Total		75	75	75	75	300

RYW = read-your-writes; MR = monotonic reads; MW = monotonic writes; WFR = writes-follow-reads. Each model was tested 75 times across the three scenarios and five configurations.

How the scenarios were applied

Normal operation: all three nodes were running and connected. Node failure: n3 was killed using SIGKILL, and test requests used surviving nodes. Network partition: n3 was isolated from {n1, n2} for replication traffic while client connections remained available.

The main node-failure and partition scenarios were each established once and used for 100 trials. The whole cluster was not restarted, and the fault was not independently reinjected before every repetition.

What each consistency test checked

Test	Check performed
Read-your-writes	Whether a read sees the client's previously acknowledged write.
Monotonic reads	Whether a later read returns an older value than an earlier read.
Monotonic writes	Whether a successor write becomes visible while its predecessor is absent.
Writes-follow-reads	Whether a write made after reading a dependency becomes visible without that dependency.

Additional experiments

Experiment	Repetitions	Observed result
BLOCKING read repair; changing the participating read quorum	5	1 → 1: 5 trials
NONE read repair; changing the participating read quorum	5	1 → 0: 5 trials
Decreasing timestamp control at ALL/ALL	1	Both writes were acknowledged; the value with the higher timestamp remained visible.

The BLOCKING control preserved the value in all five trials. The NONE control showed a 1-to-0 monotonic-read violation in all five trials. The timestamp control was performed once and consisted of two writes followed by one read, after initialization.

Main-trial outcomes and counting rules

Outcome	Trials
Violation witnesses	29
No violation observed	179
Inconclusive	92
Total main trials	300

Inconclusive trials are included in the repetition counts. Their failed operations were not retried. An unavailable or timed-out operation is not counted as a consistency violation. A completed trial without a witness does not prove that the database always provides that guarantee.

The 300 main trials contained 900 database operations. Trial totals exclude initialization, startup checks, recovery operations and software unit tests. The read-repair and timestamp controls are additional to the main matrix.

Evidence: results/20260912T063741Z/trials.json, read_repair.json and timestamp_control.json. This summary was generated from the saved experiment records. See [Cassandra_Consistency_Report.pdf](#) for the full methodology, analysis, references and AI-use disclosure.